General principle of chemistry I

[illegible] to Arrhenius, Bronsted-Lowry & Lewis concept of Acid & Base

* Acid & base are important substances across chemistry and biology.
* They participate in many reactions through proton transfer.
* The unique properties of acids and base have useful applications.

Comparing Acid-Base Theories

Theory	Acid Definition	Base Definition
Arrhenius	H^+ producer	OH^- producer
Bronsted-Lowry	Proton donor	Proton acceptor
Lewis	Electron acceptor	Electron donor

Overlaps:

(i) All [illegible] proton transfer reactions

(ii) Relative acid/base strengths.

Arrhenius Concepts of Acid & Base

⟹ An acid are substance that dissociates in water to produce (H^+) ion or hydroxonium (H_3O^+) ion as it's only positive ion.

Example: $HCl \longrightarrow H^+ + Cl^-$

$H_2SO_4 \longrightarrow 2H^+ + SO_4^-$

⟹ Base is defined as a substance that dissociates in water to produce hydroxide ion (OH^-) as it's only negative ion.

Example: $NaOH \longrightarrow Na^+ + OH^-$

$KOH \longrightarrow K^+ + OH^-$

* Furthermore both acid & bases can be classified as strong or weak acid or base depending on the hydrogen (H^+) or hydroxide (OH^-) concentration produced in solution.

The advantages of the modification of Arrhenius theory are;

(i) It recognise the role of water as a solvent in dissociation of an acid.

(ii) It also explain why substance such as hydrogen chloride and ethanoic acid can only show their usual acidic properties in the presence of water.

⇒ According to Arrhenius theory: Bases are substance which react with H^+ ion to produce or form water.

Example: $CuO + 2H^+ \longrightarrow Cu^{2+} + \boxed{H_2O}$

NOTE: Soluble base are known as alkali and it dissociate in water to produce OH^-.

Example: $NaOH \longrightarrow Na^+ + \boxed{OH^-}$

$NH_4OH \longrightarrow NH_4^+ + OH^-$

Bronsted-Lowry Concept of Acid & Base

⇒ An acid is defined as a substance that donates a proton or gives out a proton to another compound.

⇒ Base is defined as a substance that accepts a proton from another compound.

Example: $\underset{\text{donor}}{\boxed{HCl}} + \underset{\text{acceptor}}{\boxed{H_2O}} \longrightarrow \underset{\text{acceptor}}{\boxed{H_3O^+}} + \underset{\text{donor}}{\boxed{Cl^-}}$

※ A proton is a nuclear particle with a unit positive electrical charge. It is represented by the symbol H^+ because it constitutes the nucleus of a hydrogen atom.

According to Bronsted-Lowry: a substance can function as an acid only in the presence of a base. Similarly, a substance can only function as a base only in the presence of an acid. When an acidic substance loses a proton, it forms a base ~~substance~~ called the conjugate base of an acid (conjugate base). And when a basic substance gains a proton, it forms an acid called the conjugate acid of a base (conjugate acid). Thus, the reaction between an acidic substance such as hydrochloric acid, and a base:

- Substance such as ammonia (NH_3) may be represented by the equation:

$$HCl + NH_3 \longrightarrow \underbrace{\boxed{NH_4^+}}_{\text{conjugate acid}} + \boxed{Cl^-} \text{ conjugate base}$$

⟹ In the equation above the ammonium ion (NH_4^+) is the acid conjugate to the base while chloride ion (Cl^-) is the base conjugate to the acid.

Relative Strength of Acid & Base

Acids differ in their ability to donate protons to water in aqueous solution. Strong acids donate almost all their protons to water molecule so that the equilibrium is well over to the conjugate base. A strong acid is therefore weak. If an acid is weak its conjugate base is strong since it must have a strong affinity for protons. For example ethanoate ion is a strong base.

Examples

Name	Formula	Acid	Base	Name of Base
Ethanol	C_2H_5OH	H^+ +	$C_2H_5O^-$	Ethoxide
Water	H_2O	H^+ +	OH^-	Hydroxide
Ammonium	NH_4^+	H^+ +	NH_3	Ammonia
Hydrogen Sulphide	H_2S	$2H^+$ +	S^{2-}	Sulphide
Ethanoic acid	CH_3COOH	H^+ +	CH_3COO^-	Ethanoate

Lewis Concept of Acid & Base

According to Lewis, an acid is an electron pair acceptor while a base is an electron pair donor. This satisfies the requirement of the Arrhenius & Bronsted-Lowry concepts. It also involves other substance not covered by these concepts. For example, HCl is an Arrhenius acid or Bronsted-Lowry acid as well as Lewis's acid. But $AlCl_3$ is Lewis acid, it is not an Arrhenius acid or Bronsted-Lowry acid.

Example

$$SnCl_4 + 2Cl^- \longrightarrow SnCl_6^{2-}$$

$SnCl_4$ accepts 2 electron pair of Cl^- ions. Therefore, $SnCl_4$ is the Lewis acid and Cl^- is the Lewis base.

$$Hg^{2+}_{(aq)} + 4CN^{-}_{(aq)} \longrightarrow Hg(CN)_{2}^{4-}{}_{(aq)}$$

Hg^{2+} accepts 4 pairs of electrons from CN^- ions. Therefore Hg^{2+} is the Lewis acid and CN^- is the Lewis base.

Examples: Classify each of the specie as Lewis acid or Lewis base.

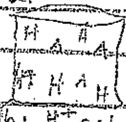
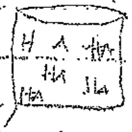
(i) CO_2 (v) NH_3
(ii) H_2O (vi) H^-
(iii) I^- (vii) H^+
(iv) SO_2 (viii) BCl_3

Answer:

i) Lewis acid (i) CO_2 ii) Lewis base (i) H_2O
(ii) SO_2 (ii) I^-
(iii) H^+ (iii) NH_3
(iv) BCl_3 (iv) H^-

=> Weak & Strong Acids

A weak acid is an acid that is partially dissociated into its ions in an aqueous solution or water. In contrast, a strong acid are fully dissociated into its ions in water. Some examples of:

Strong acids | Weak acids
(i) HCl (hydrochloric acid)  (i) CH_3COOH (ethanoic acid) 
(ii) H_2SO_4 (sulphuric acid) (ii) C_6H_5COOH (Benzoic acid)

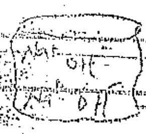
all the H^+ get dissociated

Just few were dissociated.

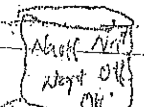
=> Strong & weak bases

A strong base is defined as chemical substance that get dissociated completely in an aqueous solution. In contrast, a weak base is a base that does not dissociate (ionize) completely in an aqueous solution. Some of the base does not bond with the hydrogen ions (H^+) from the water. A weak base is only weakly or partially ionised in water.

Strong base | Weak base
1. KOH (Potassium hydroxide) | 1. NH_3 (ammonia)
2. NaOH (Sodium hydroxide) | 2. CH_3NH_2 (Methyl amine)



completely dissociated



partially dissociated

Dissociation Constant

Dissociation constant is defined as the ratio of dissociated ions (products) to the original acid or base in the solution (reactants). For acids the acid dissociation constant (K_a) is defined as the ratio of concentration of dissociated ions to the concentration of the original acid.

⇒ K_a is a measure of how much an acid dissociates in solution, essentially indicating its strength; a higher K_a value means a stronger acid, as it readily donates hydrogen ions (H^+) when dissolved in water.

$$HA \longrightarrow H^+ + A^-$$

And $K_a = \dfrac{[H^+][A^-]}{[HA]}$ (acid dissociation constant)

* Ostwald's Dilution Law

Ostwald's dilution law refers to the electrolyte's dissociation constant with respect to the degree of dissociation. The law states that "The degree of dissociation of a weak electrolyte is inversely proportional to the square root of molar concentration or directly proportional to the square root of volume holding one mole of the solute for a weak electrolyte". This applies only to weak electrolytes such as CH_3COOH (acetic acid). The equation of Ostwald's dilution law is given as follows:

$$K_a = \frac{\alpha^2 c}{(1-\alpha)}$$

Limitations of Ostwald's Dilution law

* Oswald's dilution law is applicable only to weak electrolytes, while it is completely not applicable to strong electrolytes.
* Oswald's dilution law is based on Arrhenius theory. Thus, only some fraction of electrolytes are dissociated at normal dilutions but it is completely dissociated in infinite dilution. The strong electrolytes are completely dissociated at all dilutions. Therefore does not give an accurate value of the degree of dissociation.
* The law has applied the law of mass action, thus, a simple form of the law of mass action can not be applied in such a case.

* The ions that are dissociated might be hydrated by the water molecules, which can alter the ion's concentration.

Ionic Product of Water

Water undergoes self-ionization to form oxonium ions (hydroxonium ions).

$$2H_2O \longrightarrow H_3O^+ + OH^-$$

* The ionic product of water (Kw) is a measure of the concentration of hydrogen ions (H^+) and hydroxide ions (OH^-) in water.

⇒ Kw represent the equilibrium constant of water. And thus Kw is called ionic product of water. It has a unit of $(\text{conc})^2$ i.e $mol/dm^3 \times mol/dm^3$. The exact value depends on temperature.

At 25°c the ionic product of water (Kw) is 1.0×10^{-14}

$K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$ (at 25°c).

This means that in pure water, the concentration of hydrogen ions and hydroxide ions are equal, and the pH is neutral (pH 7).

Importance; It is essential in acid-base, reactions, aqueous solution and chemistry.

⇒ There is a relationship between ionic product (Kw) water & acid & base dissociation constant.

Example: Consider the dissociation of ethanoic acid in water as;

$$CH_3COOH + H_2O \longrightarrow H_3O^+ + CH_3COO^-_{(aq)}$$

$$\therefore K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} \text{ ------- eqn (i)}$$

for conjugate base CH_3COO^- when dissolved in water.

$$CH_3COO^- + H_2O \longrightarrow CH_3COOH + OH^-$$

$$\therefore K_b = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]} \text{ ——— (ii)}$$

Multiply equation (i) by (ii) we have

$$K_a \times K_b = [H_3O^+][OH^-] \text{ ——— (iii)}$$

$$K_a \times K_b = K_w \text{ ——— (iv)}$$

Taking the logarithm of the equation (iv) we have
$pK_a + pK_b = pK_w$ .... (v)

At 25°C $pK_b = 14$
$\therefore pH + pOH = 14$ .... (vi)
from the equation (v) pK_a is the value for H^+ ions (acids) &
pK_b is the value of its conjugate base.

* pH
In chemistry, pH (potential of Hydrogen) is a measure of the concentration of hydrogen ions (H^+) in a solution. It's a scale used to express the acidity or basicity (alkalinity) of a solution.

pH Scale:
- pH 0 – 1 : Strongly acidic
- pH 2–3 : Acidic
- " 4–5 : Weakly acidic
- " 6–7 : Neutral (neither acidic nor basic).

pH Calculation
$pH = -\log[H^+]$
where $[H^+]$ is the concentration of hydrogen ions in moles per liter (M)

Importance of pH:
pH plays a crucial role in various chemical, biological & environmental processes, such as
- Chemical reaction & equilibria.
- Biological systems & enzyme activity
- Environmental monitoring & water quality assessment.

While a pH of a base is termed pOH (potential of hydroxide) and is defined as a measure of the concentration of hydroxide ions (OH^-) in a solution.

pOH Scale:
- pOH 0 – 1 : Strongly basic
- " 2–3 : Basic
- " 4–5 : Weakly basic

pOH 6 – 7 : Neutral (neither acidic nor basic)
" 8 – 9 : weakly acidic
" 10 – 12 : Acidic
pOH 13 – 14 : Strongly acidic

pOH Calculation:
$pOH = -\log[OH^-]$

Relationship between pH & pOH:
$pH + pOH = 14$ (at 25°C)

⇒ This means that if you know the pH of a solution, you can calculate the pOH and vice versa.

Examples:
1. Calculate the pH of
(a) 1.0×10^{-3} m of HCl
(b) 0.020m $Ba(OH)_2$ solution

Answers
(a) HCl will completely ionize because it is a strong acid

$$HCl \longrightarrow H^+ + Cl^-$$

	HCl	H^+	Cl^-
Initial	1.0×10^{-3} m	0.00m	0.00m
change	-1.0×10^{-3} m	1.0×10^{-3} m	1.0×10^{-3} m
final	0.00m	1.0×10^{-3} m	1.0×10^{-3} m

Where (+) charge indicate increase in concentration and (–) charge indicate decrease in concentration

$\therefore H^+ = 1.0 \times 10^{-3}$

$pH = -\log[H^+]$
$= -\log[1.0 \times 10^{-3}]$
$= -[-3]$
$= 3$

The pH of 1.0×10^{-3} m of HCl is equal to $= 3$

(b) $Ba(OH)_2$ is a strong base which ionises as follows

ANSWER:

$$Ba(OH)_2 \longrightarrow Ba^{2+} + 2OH^-$$

	$Ba(OH)_2$	Ba^{2+}	$2OH^-$
I	0.020m	0.00m	0.00m
C	−0.020m	0.020m	2×0.020m
E	0.00m	0.020m	0.04m

$\therefore [OH^-] = 0.04m$

$pOH = -\log[0.04] \Rightarrow 1.4$

Therefore $pH = 14 - pOH$

$pH = 14 - 1.4$

$pH = 12.60$ //

(c) Calculate the pH & the pOH of 0.06mol/dm³ of H_2SO_4 Solution.

ANSWER:

$$H_2SO_4 \quad 2H^+ + SO_4^{2-}$$

0.06mol/dm³ 2×0.06mol/dm³ 0.06mol/dm³

$H^+ = 2 \times 0.06 = 0.12 mol/dm^3$

$pH = -\log[H^+] \Rightarrow pH = -\log(0.12)$

$pH = 0.92$

And for the pOH

$pOH = 14 - pH$

$= 14 - 0.92$

$pOH = 13.08$ //